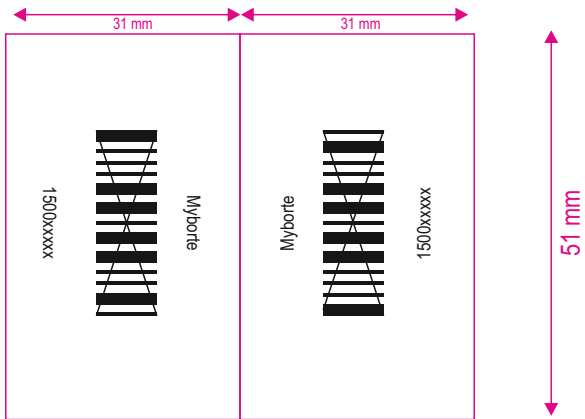


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Dr.Reddy's
Rx Prescription Drug

For the use of an Oncologist or a Hospital or a Cancer Institutions only
MYBORTE 3.5mg
(Bortezomib Powder for Solution for Injection 3.5mg/vial)

COMPOSITION

Each vial contains: Bortezomib 3.5 mg (lyophilized)

Excipients:

Manitol (Pyrogen free), Tertiary Butanol, Water for Injection, Nitrogen NF (Process Aid)

Product Description

Bortezomib for Injection 3.5 mg is a White to Off White Cake or Powder filled in 10 mL single-use USP Type-I tubular glass vial stoppered with 13 mm two leg slotted bromobutyl rubber stopper and sealed with 13 mm aluminium flip-off seal.
Bortezomib Powder for Solution for Injection 3.5 mg/vial should be reconstituted with 0.9% Sodium Chloride Injection before administration.
Description of Reconstituted solution: Clear colorless solution free from visible extraneous matter.

CLINICAL INFORMATION

Therapeutic Indications

"Bortezomib is indicated for the treatment of patients with multiple myeloma. Bortezomib is indicated for the treatment of patients with mantle cell lymphoma who have received at least one prior therapy."

Recommended dose

"General Dosing Guidelines" The recommended starting dose of bortezomib is 1.3 mg/m². Bortezomib may be administered intravenously at a concentration of 1 mg/mL, or subcutaneously at a concentration of 2.5 mg/mL. When administered intravenously, Bortezomib is administered as a 3 to 5 second bolus intravenous injection. Bortezomib is for intravenous or subcutaneous use only. Bortezomib should not be administered by any other route. Because each route of administration has a different reconstituted concentration, caution should be used when calculating the volume to be administered.

BORTEZOMIB IS FOR INTRAVENOUS OR SUBCUTANEOUS USE ONLY. Intrathecal administration has resulted in death. Dosage in Previously Untreated Multiple Myeloma Bortezomib is administered in combination with oral melphalan and oral prednisone for nine 6-week treatment cycles as shown in Table 1.

In Cycles 1-4, bortezomib is administered twice weekly (days 1, 4, 8, 11, 22, 25, 29 and 32).

In Cycles 5-9, bortezomib is administered once weekly (days 1, 8, 22 and 29). At least 72 hours should elapse between consecutive doses of bortezomib.

Table 1: Dosage Regimen for Patients with Previously Untreated Multiple Myeloma

Twice weekly bortezomib (cycles 1-4)											
Week	1				2		3		4		5
Bortezomib (1.3 mg/m ²)	Day 1	--	--	Day 4	Day 8	Day 11	rest period	Day 22	Day 25	Day 29	Day 32
Melphalan (9 mg/m ²)	Day 1	Day 2	Day 3	Day 4	--	--	rest period	--	--	--	--
Prednisone (60 mg/m ²)							rest period				rest period
Once weekly bortezomib (cycles 5-9)											
Week	1				2		3		4		5
Bortezomib (1.3 mg/m ²)	Day 1	--	--	--	Day 8		rest period		Day 22		Day 29
Melphalan (9 mg/m ²)	Day 1	Day 2	Day 3	Day 4	--		rest period	--	--		rest period
Prednisone (60 mg/m ²)							rest period				rest period

Dose Modification Guidelines for Combination Therapy with bortezomib, Melphalan and Prednisone: Prior to initiating any cycle of therapy with bortezomib in combination with melphalan and prednisone, Platelet count should be $\geq 70 \times 10^9/L$ and the ANC should be $\geq 1 \times 10^9/L$. nonhematological toxicities should have resolved to Grade 1 or baseline.

Table 2: Dose modifications during cycles of combination of bortezomib, Melphalan and Prednisone therapy.

Toxicity	Dose modification or delay
Hematological toxicity during a cycle:	
If prolonged Grade 4 neutropenia or thrombocytopenia, or thrombocytopenia with bleeding is observed in the previous cycle	Consider reduction of the melphalan dose by 25% in the next cycle.
If platelet counts $\leq 30 \times 10^9/L$ or ANC $\leq 0.75 \times 10^9/L$ on a bortezomib dosing day (other than day 1)	Withhold bortezomib dose
If several bortezomib doses in consecutive cycles are withheld due to toxicity	bortezomib dose should be reduced by 1 dose level (from 1.3 - 1 mg/m ² , or from 1 - 0.7 mg/m ²)
Grade ≥ 3 non-hematological toxicities	Bortezomib therapy should be withheld until symptoms of the toxicity have resolved to Grade 1 or baseline. Then, bortezomib may be reinitiated with one dose level reduction (from 1.3 - 1 mg/m ² , or from 1 - 0.7 mg/m ²). For bortezomib-related neuropathic pain and/or peripheral neuropathy, hold and/or modify bortezomib as outlined in Table 3.

For information concerning melphalan and prednisone, see manufacturer's prescribing information.

Dose modifications guidelines for peripheral neuropathy are provided.

Dosage and Dose Modifications for Relapsed Multiple Myeloma and Relapsed Mantle Cell Lymphoma:

Bortezomib (1.3 mg/m²/dose) is administered twice weekly for 2 weeks (Days 1, 4, 8, and 11) followed by a 10-day rest period (Days 12 - 21). For extended therapy of more than 8 cycles, bortezomib may be administered on the standard schedule or, for relapsed multiple myeloma, on a maintenance schedule of once weekly for 4 weeks (Days 1, 8, 15, and 22) followed by a 13-day rest period (Days 23 to 35). At least 72 hours should elapse between consecutive doses of bortezomib.

Bortezomib therapy should be withheld at the onset of any Grade 3 non hematological or Grade 4 hematological toxicities excluding neuropathy as discussed below. Once the symptoms of the toxicity have resolved, bortezomib therapy may be reinitiated at a 25% reduced dose (1.3 mg/m² dose reduced to 1.0 mg/m² /dose; 1.0 mg/m² dose reduced to 0.7 mg/m² /dose).

For dose modifications guidelines for peripheral neuropathy, see Dose modifications for peripheral neuropathy below.

Dose Modifications for Peripheral Neuropathy: Starting bortezomib subcutaneously may be considered for patients with pre-existing or at high risk of peripheral neuropathy. Patients with pre-existing severe neuropathy should be treated with bortezomib only after careful risk-benefit assessment. Patients experiencing new or worsening peripheral neuropathy during bortezomib therapy may require a decrease in the dose and/or a less dose-intensive schedule. For dose or schedule modification guidelines for patients who experience bortezomib-related neuropathic pain and/or peripheral neuropathy see Table 3.

Table 3: Recommended dose modification for bortezomib-related neuropathic pain and/or Peripheral sensory or motor Neuropathy

Grade 1 (Paresthesias and/or loss of reflexes) without pain or loss of function	No action
Grade 1 with pain or Grade 2 (interfering with function but not with activities of daily living)	Reduce bortezomib to 1mg/m ²
Grade 2 with pain or grade 3 (interfering with activities of daily living)	Withhold bortezomib therapy until toxicity resolves. When toxicity resolves reinitiate with a reduced dose of bortezomib at 0.7 mg/m ² and change treatment schedule to once per week.
Grade 4 (Sensory neuropathy which is disabling or motor neuropathy that is life - threatening or leads to paralysis)	Discontinue bortezomib
Grading based on NCI common toxicity Criteria CTCAE v3.0.	

Special populations

Elderly

There is no evidence to suggest that dose adjustments are necessary in patients over 65 years of age.

There are no studies on the use of bortezomib in elderly patients with previously untreated multiple myeloma who are eligible for high-dose chemotherapy with haematopoietic stem cell transplantation. Therefore no dose recommendations can be made in this population.

Previously untreated mantle cell lymphoma patients, who exposed to bortezomib were in the range 65-74 years and ≥ 75 years of age, respectively. In patients aged ≥ 75 years, both regimens, Bortezomib-R-CAP as well as R-CHOP, were less tolerated

Hepatic impairment

Patients with mild hepatic impairment do not require a dose adjustment and should be treated per the recommended dose. Patients with moderate or severe hepatic impairment should be started on bortezomib at a reduced dose of 0.7 mg/m² per injection during the first treatment cycle, and a subsequent dose escalation to 1.0 mg/m² or further dose reduction to 0.5 mg/m² may be considered based on patient tolerability (see Table 4).

Table 4: Recommended starting dose modification for bortezomib in patients with hepatic impairment

Grade of hepatic impairment*	Bilirubin level	SGOT (AST) levels	Modification of starting dose
Mild	$\leq 1.0 \times$ ULN	$> ULN$	None
	$> 1.0 \times -1.5 \times$ ULN	Any	None
Moderate	$> 1.5 \times -3 \times$ ULN	Any	Reduce bortezomib to 0.7 mg/m ² in the first treatment cycle. Consider dose escalation to 1.0 mg/m ² or further dose reduction to 0.5 mg/m ² in subsequent cycles based on patient tolerability.
Severe	$> 3 \times$ ULN	Any	

Abbreviations: SGOT=serum glutamic oxaloacetic transaminase; AST=aspartate aminotransferase; ULN=upper limit of the normal range.

* Based on NCI Organ Dysfunction Working Group classification for categorizing hepatic impairment (mild, moderate, severe).

Renal impairment

The pharmacokinetics of bortezomib are not influenced in patients with mild to moderate renal impairment (Creatinine Clearance [CrCL] > 20 mL/min/1.73 m²); therefore, dose adjustments are not necessary for these patients. It is unknown if the pharmacokinetics of bortezomib are influenced in patients with severe renal impairment not undergoing dialysis (CrCL < 20 mL/min/1.73 m²).

Since dialysis may reduce bortezomib concentrations, bortezomib should be administered after the dialysis procedure.

Paediatric population

The safety and efficacy of bortezomib in children below 18 years of age have not been established.

Method of administration

Bortezomib 3.5 mg powder for solution for injection is available for intravenous or subcutaneous administration.

Bortezomib 1 mg powder for solution for injection is available for intravenous administration only.

Bortezomib should not be given by other routes. Intrathecal administration has resulted in death.

Intravenous injection

Bortezomib 3.5 mg reconstituted solution is administered as a 3-5 second bolus intravenous injection through a peripheral or central intravenous catheter followed by a flush with sodium chloride 9 mg/mL (0.9%) solution for injection. At least 72 hours should elapse between consecutive doses of bortezomib.

Subcutaneous injection

Bortezomib 3.5 mg reconstituted solution is administered subcutaneously through the thighs (right or left) or abdomen (right or left). The solution should be injected subcutaneously, at a 45-90° angle. Injection sites should be rotated for successive injections.

If local injection site reactions occur following bortezomib subcutaneous injection, either a less concentrated bortezomib solution (bortezomib 3.5 mg to be reconstituted to 1 mg/mL instead of 2.5 mg/mL) may be administered subcutaneously or a switch to intravenous injection is recommended.

Special precautions for disposal and other handling

General precautions

Bortezomib is a cytotoxic agent. Therefore, caution should be used during handling and preparation of bortezomib. Use of gloves and other protective clothing to prevent skin contact is recommended.

Aseptic technique must be strictly observed throughout the handling of bortezomib, since it contains no preservative.

There have been fatal cases of inadvertent intrathecal administration of bortezomib. Bortezomib 1 mg powder for solution for injection is for intravenous use only, while bortezomib 3.5 mg powder for solution for injection is for intravenous or subcutaneous use. Bortezomib should not be administered intrathecally.

Instructions for reconstitution

Bortezomib must be reconstituted by a healthcare professional.

Intravenous injection

Each 10 mL vial of bortezomib must be reconstituted with 3.5 mL of sodium chloride 9 mg/mL (0.9%) solution for injection. Dissolution of the lyophilized powder is completed in less than 2 minutes.

After reconstitution, each mL solution contains 1 mg bortezomib. The reconstituted solution is clear and colorless, with a final pH of 4 to 7.

The reconstituted solution must be inspected visually for particulate matter and discoloration prior to administration. If any discoloration or particulate matter is observed, the reconstituted solution must be discarded.

Subcutaneous injection

Each 10 mL vial of bortezomib must be reconstituted with 1.4 mL of sodium chloride 9 mg/mL (0.9%) solution for injection. Dissolution of the lyophilized powder is completed in less than 2 minutes.

After reconstitution, each mL solution contains 2.5 mg bortezomib. The reconstituted solution is clear and colorless, with a final pH of 4 to 7.

The reconstituted solution must be inspected visually for particulate matter and discoloration prior to administration. If any discoloration or particulate matter is observed, the reconstituted solution must be discarded.

Disposal

Bortezomib is for single use only. Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

Contraindications

Hypersensitivity to the active substance, to any of the excipients. Acute diffuse infiltrative pulmonary and pericardial disease.

Special Warnings and Precautions for Use

When thalidomide is used, particular attention to pregnancy testing and prevention requirements is needed.

Intrathecal administration

There have been fatal cases of inadvertent intrathecal administration of bortezomib. Bortezomib 1 mg powder for solution for injection is for intravenous use only, while bortezomib 3.5 mg powder for solution for injection is for intravenous or subcutaneous use. Bortezomib should not be administered intrathecally.

Gastrointestinal toxicity

Gastrointestinal toxicity, including nausea, diarrhoea, vomiting and constipation are very common with bortezomib treatment. Cases of ileus have been uncommonly reported. Therefore, patients who experience constipation should be closely monitored.

Haematological toxicity

Bortezomib treatment is very commonly associated with haematological toxicities (thrombocytopenia, neutropenia and anaemia). Bortezomib (injected intravenously) versus dexamethasone, the most common haematologic toxicity was transient thrombocytopenia. Platelets were lowest at day 11 of each cycle of bortezomib treatment. There was no evidence of cumulative thrombocytopenia. The mean platelet count nadir measured was approximately with baseline. In patients with advanced myeloma the severity of thrombocytopenia was related to pre-treatment platelet count: for baseline platelet counts $< 75,000/\mu L$ and $\leq 25,000/\mu L$.

There was a higher incidence of Grade ≥ 3 thrombocytopenia in the bortezomib treatment group (Bortezomib-R-CAP) as compared to the non-bortezomib treatment group (rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisone [R-CHOP]). The two treatment groups were similar with regard to the overall incidence of all-grade bleeding events as well as Grade 3 and higher bleeding events. In the Bortezomib-R-CAP group, patients received platelet transfusions compared to lower in patients with R-CHOP group.

Gastrointestinal and intracerebral haemorrhage, have been reported in association with bortezomib treatment. Therefore, platelet counts should be monitored prior to each dose of bortezomib. Bortezomib therapy should be withheld when the platelet count is $< 25,000/\mu L$ or, in the case of combination with melphalan and prednisone, when the platelet count is $\leq 30,000/\mu L$. Potential benefit of the treatment should be carefully weighed against the risks, particularly in case of moderate to severe thrombocytopenia and risk factors for bleeding.

Complete blood counts (CBC) with differential and including platelet counts should be frequently monitored throughout treatment with bortezomib. Platelet transfusion should be considered when clinically appropriate.

Transient neutropenia that was reversible between cycles was observed, with no evidence of cumulative neutropenia. Neutrophils were lowest at Day 11 of each cycle of bortezomib treatment and typically recovered to baseline by the next cycle. Patients with neutropenia are at increased risk of infections; they should be monitored for signs and symptoms of infection and treated promptly. Granulocyte colony stimulating factors may be administered for haematologic toxicity according to local standard practice. Prophylactic use of granulocyte colony stimulating factors should be considered in case of repeated delays in cycle administration.

Herpes zoster virus reactivation

Antiviral prophylaxis should be considered in patients being treated with bortezomib. Patients with previously untreated multiple myeloma, the overall incidence of herpes zoster reactivation was more common in patients treated with bortezomib + Melphalan + Prednisone compared with Melphalan + Prednisone. The incidence of herpes zoster infection was higher in the Bortezomib-R-CAP arm and lower in the R-CHOP arm.

Hepatitis B Virus (HBV) reactivation and infection

When rituximab is used in combination with bortezomib, HBV screening must always be performed in patients at risk of infection with HBV before initiation of treatment. Carriers of hepatitis B and patients with a history of hepatitis B must be closely monitored for clinical and laboratory signs of active HBV infection during and following rituximab combination treatment with bortezomib.

Antiviral prophylaxis should be considered

Progressive multifocal leukoencephalopathy (PML)

Very rare cases with unknown causality of John Cunningham (JC) virus infection, resulting in PML and death, have been reported in patients treated with bortezomib. Patients diagnosed with PML had prior or concurrent immunosuppressive therapy. Most cases of PML were diagnosed within 12 months of their first dose of bortezomib. Patients should be monitored at regular intervals for any new or worsening neurological symptoms or signs that may be suggestive of PML as part of the differential diagnosis of CNS problems. If a diagnosis of PML is suspected, patients should be referred to a specialist in PML and appropriate diagnostic measures for PML should be initiated. Discontinue bortezomib if PML is diagnosed.

Peripheral neuropathy

Treatment with bortezomib is very commonly associated with peripheral neuropathy, which is predominantly sensory. However, cases of severe motor neuropathy with or without sensory peripheral neuropathy have been reported. The incidence of peripheral neuropathy increases early in the treatment and has been observed to peak during cycle 5. It is recommended that patients be carefully monitored for symptoms of neuropathy such as a burning sensation, hyperesthesia, hypoaesthesia, paraesthesia, discomfort, neuropathic pain or weakness.

Bortezomib administered intravenously versus subcutaneously, the incidence of Grade ≥ 2 peripheral neuropathy events lower for the subcutaneous injection group and higher for the intravenous injection group. Grade ≥ 3 peripheral neuropathy occurred in less patients in the subcutaneous treatment group, compared with intravenous treatment group. The incidence of all grade peripheral neuropathy with bortezomib administered intravenously was lower.

Patients experiencing new or worsening peripheral neuropathy should undergo neurological evaluation and may require a change in the dose, schedule or route of administration to subcutaneous. Neuropathy has been managed with supportive care and other therapies.

Early and regular monitoring for symptoms of treatment-emergent neuropathy with neurological evaluation should be considered in patients receiving bortezomib in combination with medicinal products known to be associated with neuropathy (e.g. thalidomide) and appropriate dose reduction or treatment discontinuation should be considered.

In addition to peripheral neuropathy, there may be a contribution of autonomic neuropathy to some adverse reactions such as postural hypotension and severe constipation with ileus. Information on autonomic neuropathy and its contribution to these undesirable effects is limited.

Seizures

Seizures have been uncommonly reported in patients without previous history of seizures or epilepsy. Special care is required when treating patients with any risk factors for seizures.

Hypotension

Bortezomib treatment is commonly associated with orthostatic/postural hypotension. Most adverse reactions are mild to moderate in nature and are observed throughout treatment. Patients who developed orthostatic hypotension on bortezomib (injected intravenously) did not have evidence of orthostatic hypotension prior to treatment with bortezomib. Most patients required treatment for their orthostatic hypotension. A minority of patients with orthostatic hypotension experienced syncope events. Orthostatic/postural hypotension was not acutely related to bolus infusion of bortezomib. The mechanism of this event is unknown although a component may be due to autonomic neuropathy. Autonomic neuropathy may be related to bortezomib or bortezomib may aggravate an underlying condition such as diabetic or amyloidotic neuropathy. Caution is advised when treating patients with a history of syncope receiving medicinal products known to be associated with hypotension; or who are dehydrated due to recurrent diarrhoea or vomiting. Management of orthostatic/postural hypotension may include adjustment of antihypertensive medicinal products, rehydration or administration of mineralocorticosteroids and/or sympathomimetics. Patients should be instructed to seek medical advice if they experience symptoms of dizziness, light-headedness or fainting spells.

Posterior Reversible Encephalopathy Syndrome (PRES)

There have been reports of PRES in patients receiving bortezomib. PRES is a rare, often reversible, rapidly evolving neurological condition, which can present with seizure, hypertension, headache, lethargy, confusion, blindness, and other visual and neurological disturbances. Brain imaging, preferably Magnetic Resonance Imaging (MRI), is used to confirm the diagnosis. In patients developing PRES, bortezomib should be discontinued.

Heart failure

Acute development or exacerbation of congestive heart failure, and/or new onset of decreased left ventricular ejection fraction has been reported during bortezomib treatment. Fluid retention may be a predisposing factor for signs and symptoms of heart failure. Patients with risk factors for or existing heart disease should be closely monitored.

Electrocardiogram investigations

There have been isolated cases of QT-interval prolongation in clinical studies, causality has not been established.

Pulmonary disorders

There have been rare reports of acute diffuse infiltrative pulmonary disease of unknown aetiology such as pneumonitis, interstitial pneumonia, lung infiltration, and acute respiratory distress syndrome (ARDS) in patients receiving bortezomib. Some of these events have been fatal. A pre-treatment chest radiograph is recommended to serve as a baseline for potential post-treatment pulmonary changes.

In the event of new or worsening pulmonary symptoms (e.g., cough, dyspnoea), a prompt diagnostic evaluation should be performed and patients treated appropriately. The benefit/risk ratio should be considered prior to continuing bortezomib therapy.

Damunonib and bortezomib with concomitant administration of high-dose cytarabine (2 g/m² per day) by continuous infusion over 24 hours is not recommended.

Renal impairment

Renal complications are frequent in patients with multiple myeloma. Patients with renal impairment should be monitored closely.

Hepatic impairment

Bortezomib is metabolised by liver enzymes. Bortezomib exposure is increased in patients with moderate or severe hepatic impairment; these patients should be treated with bortezomib at reduced doses and closely monitored for toxicities.

Hepatic reactions

Rare cases of hepatic failure have been reported in patients receiving bortezomib and concomitant medicinal products and with serious underlying medical conditions. Other reported hepatic reactions include increases in liver enzymes, hyperbilirubinaemia, and hepatitis. Such changes may be reversible upon discontinuation of bortezomib.

Tumour lysis syndrome

Because bortezomib is a cytotoxic agent and can rapidly kill malignant plasma cells, the complications of tumour lysis syndrome may occur. The patients at risk of tumour lysis syndrome are those with high tumour burden prior to treatment. These patients should be monitored closely and appropriate precautions taken.

Concomitant medicinal products

Patients should be closely monitored when given bortezomib in combination with potent CYP3A4-inhibitors. Caution should be exercised when bortezomib is combined with CYP3A4- or CYP2C19-substrates.

Normal liver function should be confirmed and caution should be exercised in patients receiving oral hypoglycemics.

Potentially immunocomplex-mediated reactions

Potentially immunocomplex-mediated reactions, such as serum-sickness-type reaction, polyarthritides with rash and proliferative glomerulonephritis have been reported uncommonly. Bortezomib should be discontinued if serious reactions occur.

Interaction with Other Medicinal Products and Other Forms of Interaction

In vitro studies indicate that bortezomib is a weak inhibitor of the cytochrome P450 (CYP) isozymes 1A2, 2C9, 2C19, 2D6 and 3A4. Based on the limited contribution of CYP2D6 to the metabolism of bortezomib, the CYP2D6 poor metaboliser phenotype is not expected to affect the overall disposition of bortezomib.

Therefore, patients should be closely monitored when given bortezomib in combination with potent CYP3A4 inhibitors (e.g. ketoconazole, ritonavir).

The concomitant use of bortezomib with strong CYP3A4 inducers (e.g., rifampicin, carbamazepine, phenytoin, phenobarbital and St. John's Wort) is not recommended, as efficacy may be reduced.

Patients on oral antidiabetic agents receiving bortezomib treatment may require close monitoring of their blood glucose levels and adjustment of the dose of their antidiabetics.

Fertility, pregnancy and lactation

Contraception in males and females

Male and female patients of childbearing potential must use effective contraceptive measures during and for 3 months following treatment.

Pregnancy

No clinical data are available for bortezomib with regard to exposure during pregnancy. The teratogenic potential of bortezomib has not been fully investigated.

In non-clinical studies, bortezomib had no effects on embryonal/foetal development in rats and rabbits at the highest maternally tolerated doses. Animal studies to determine the effects of bortezomib on parturition and post-natal development were not conducted. Bortezomib should not be used during pregnancy unless the clinical condition of the woman requires treatment with bortezomib.

If bortezomib is used during pregnancy, or if the patient becomes pregnant while receiving this medicinal product, the patient should be informed of potential for hazard to the foetus.

Thalidomide is a known human teratogenic active substance that causes severe life-threatening birth defects. Thalidomide is contraindicated during pregnancy and in women of childbearing potential unless all the conditions of the thalidomide pregnancy prevention program are met. Patients receiving bortezomib in combination with thalidomide should adhere to the pregnancy prevention program of thalidomide. Refer to the Summary of Product Characteristics of thalidomide for additional information.

Breast-feeding

It is not known whether bortezomib is excreted in human milk. Because of the potential for serious adverse reaction in breast-fed infants, breast-feeding should be discontinued during treatment with bortezomib.

Immune system disorders	Uncommon	Angioedema*, Hypersensitivity*
	Rare	Anaphylactic shock, Amyloidosis, Type III immune complex mediated reaction
Endocrine disorders	Uncommon	Cushing's syndrome*, Hyperthyroidism*, Inappropriate antidiuretic hormone secretion
	Rare	Hypothyroidism
Metabolism and nutrition disorders	Very Common	Decreased appetite
	Common	Dehydration, Hypokalaemia*, Hyponatraemia*, Blood glucose abnormal*, Hypocalcaemia*, Enzyme abnormality*
	Uncommon	Tumour lysis syndrome, Failure to thrive*, Hypomagnesaemia*, Hypophosphataemia*, Hyperkalaemia*, Hypercalcaemia*, Hypermnatraemia*, Uric acid abnormal*, Diabetes mellitus*, Fluid retention
	Rare	Hypermagnesaemia*, Acidosis, Electrolyte imbalance*, Fluid overload, Hypochloreaemia*, Hypovolaemia, Hyperchloreaemia*, Hyperphosphataemia*, Metabolic disorder, Vitamin B complex deficiency, Vitamin B12 deficiency, Gout, Increased appetite, Alcohol intolerance
Psychiatric disorders	Common	Mood disorders and disturbances*, Anxiety disorder*, Sleep disorders and disturbances*
	Uncommon	Mental disorder*, Psychotic disorder*, Confusion*, Restlessness
	Rare	Suicidal ideation*, Adjustment disorder, Delirium, Libido decreased
Nervous system disorders	Very Common	Neuropathies*, Peripheral sensory neuropathy, Dysaesthesia*, Neuralgia*
	Common	Motor neuropathy*, Loss of consciousness (inc syncope), Dizziness*, Dysgeusia*, Lethargy, Headache*
	Uncommon	Tremor, Peripheral sensorimotor neuropathy, Dyskinesia*, Cerebellar coordination and balance disturbances*, Memory loss (exc dementia)*, Encephalopathy*, Posterior Reversible Encephalopathy Syndrome*, Neurotoxicity, Seizure disorders*, Post herpetic neuralgia, Speech disorder*, Restless legs syndrome, Migraine, Sciatica, Disturbance in attention, Reflexes abnormal*, Parosmia
	Rare	Cerebral haemorrhage*, Haemorrhage intracranial (inc subarachnoid)*, Brain oedema, Transient ischaemic attack, Coma, Autonomic nervous system imbalance, Autonomic neuropathy, Cranial palsy*, Paralysis*, Paresis*, Presyncope, Brain stem syndrome, Cerebrovascular disorder, Nerve root lesion, Psychomotor hyperactivity, Spinal cord compression, Cognitive disorder NOS, Motor dysfunction, Nervous system disorder NOS, Radiculitis, Drooling, Hypotonia
	Common	Eye swelling*, Vision abnormal*, Conjunctivitis*
	Uncommon	Eye haemorrhage*, Eyelid infection*, Eye inflammation*, Diplopia, Dry eye*, Eye irritation*, Eye pain, Lacrimation increased, Eye discharge
	Rare	Corneal lesion*, Exophthalmos, Retinitis, Scotoma, Eye disorder (inc eyelid) NOS, Dacryoadenitis acquired, Photophobia, Photopsia, Optic neuropathy*, Different degrees of visual impairment (up to blindness)*
Ear and labyrinth disorders	Common	Vertigo*
	Uncommon	Dysacusis (inc tinnitus)*,Hearing impaired (up to and inc deafness), Ear discomfort*
	Rare	Ear haemorrhage, Vestibular neuronitis, Ear disorder NOS
Cardiac disorders	Uncommon	Cardiac tamponade*, Cardio-pulmonary arrest*, Cardiac fibrillation (inc atrial), Cardiac failure (inc left and right ventricular)*, Arrhythmia*, Tachycardia*, Palpitations, Angina pectoris, Pericarditis (inc pericardial effusion)*, Cardiomyopathy*, Ventricular dysfunction*, Bradycardia
	Rare	Atrial flutter, Myocardial infarction*, Atrioventricular block*, Cardiovascular disorder (inc cardiogenic shock), Torsade de pointes, Angina unstable, Cardiac valve disorders*, Coronary artery insufficiency, Sinus arrest
Vascular disorders	Common	Hypotension*, Orthostatic hypotension, Hypertension*
	Uncommon	Cerebrovascular accident*, Deep vein thrombosis*, Haemorrhage*, Thrombophlebitis (inc superficial), Circulatory collapse (inc hypovolaemic shock), Phlebitis, Flushing*, Haematoma (inc perineal)*, Poor peripheral circulation*, Vasculitis, Hyperaemia (inc ocular)*
	Rare	Peripheral embolism, Lymphoedema, Pallor, Erythromelalgia, Vasodilatation, Vein discolouration, Venous insufficiency
Respiratory, thoracic and mediastinal disorders	Common	Dyspnoea*, Epistaxis, Upper/lower respiratory tract infection*, Cough*
	Uncommon	Pulmonary embolism, Pleural effusion, Pulmonary oedema (inc acute), Pulmonary alveolar haemorrhage*, Bronchospasm, Chronic obstructive pulmonary disease*, Hypoxaemia*, Respiratory tract congestion*, Hypoxia, Pleurisy*, Hiccups, Rhinorrhoea, Dysphonia, Wheezing
	Rare	Respiratory failure, Acute respiratory distress syndrome, Apnoea, Pneumothorax, Atelectasis, Pulmonary hypertension, Haemoptysis, Hyperventilation, Orthopnoea, Pneumonitis, Respiratory alkalosis, Tachypnoea, Pulmonary fibrosis, Bronchial disorder*, Hypocapnia*, Interstitial lung disease, Lung infiltration, Throat tightness, Dry throat, Increased upper airway secretion, Throat irritation, Upper-airway cough syndrome
Gastrointestinal disorders	Very Common	Nausea and vomiting symptoms*, Diarrhoea*, Constipation
	Common	Gastrointestinal haemorrhage (inc mucosal)*, Dyspepsia, Stomatitis*, Abdominal distension, Oropharyngeal pain*, Abdominal pain (inc gastrointestinal and splenic pain)*, Oral disorder*, Flatulence
	Uncommon	Pancreatitis (inc chronic)*, Haematemesis, Lip swelling*, Gastrointestinal obstruction (inc ileus)*, Abdominal discomfort, Oral ulceration*, Enteritis*, Gastritis*, G ingival bleeding, Gastroesophageal reflux disease*, Colitis (inc clostridium difficile)*, Colitis ischaemic*, Gastrointestinal inflammation*, Dysphagia, Irritable bowel syndrome, Gastrointestinal disorder NOS, Tongue coated, Gastrointestinal motility disorder*, Salivary gland disorder*
	Rare	Pancreatitis acute, Peritonitis*, Tongue oedema*, Ascites, Oesophagitis, Chelitis, Faecal incontinence, Anal sphincter atony, Faecaloma*, Gastrointestinal ulceration and perforation*, Gingival hypertrophy, Megacolon, Rectal discharge, Oropharyngeal blistering*, Lip pain, Periodontitis, Anal fissure, Change of bowel habit, Proctalgia, Abnormal faeces
Hepatobiliary disorders	Common	Hepatic enzyme abnormality*
	Uncommon	Hepatotoxicity (inc liver disorder), Hepatitis*, Cholestasis
	Rare	Hepatic failure, Hepatomegaly, Budd-Chiari syndrome, Cytomegalovirus hepatitis, Hepatic haemorrhage, Cholelithiasis
Skin and subcutaneous tissue disorders	Common	Rash*, Pruritus*, Erythema, Dry skin
	Uncommon	Erythema multiforme, Urticaria, Acute febrile neutrophilic dermatosis, Toxic skin eruption, Toxic epidermal necrolysis*, Stevens-Johnson syndrome*, Dermatitis*, Hair disorder*, Petechiae, Ecchymosis, Skin lesion, Purpura, Skin mass*, Psoriasis, Hyperhidrosis, Night sweats, Decubitus/ulcer*, Acne*, Blister*, Pigmentation disorder*
	Rare	Skin reaction, Jessner's lymphocytic infiltration, Palmar-plantar erythrodysaesthesia syndrome, Haemorrhage subcutaneous, Livedo reticularis, Skin induration, Papule, Photosensitivity reaction, Seborrhoea, Cold sweat, Skin disorder NOS, Erythrosis, Skin ulcer, Nail disorder
Musculoskeletal and connective tissue disorders	Very Common	Musculoskeletal pain*
	Common	Muscle spasms*, Pain in extremity, Muscular weakness
	Uncommon	Muscle twitching, Joint swelling, Arthritis*, Joint stiffness, Myopathies*,Sensation of heaviness
	Rare	Rhabdomyolysis, Temporomandibular joint syndrome, Fistula, Joint effusion, Pain in jaw, Bone disorder, Musculoskeletal and connective tissue infections and inflammations*, Synovial cyst
Renal and urinary disorders	Common	Renal impairment*
	Uncommon	Renal failure acute, Renal failure chronic*, Urinary tract infection*, Urinary tract signs and symptoms*, Haematuria*, Urinary retention, Micturition disorder*, Proteinuria, Azotaemia*, Oliguria*, Pollakiuria
	Rare	Bladder irritation
Reproductive system and breast disorders	Uncommon	Vaginal haemorrhage, Genital pain*, Erectile dysfunction,
	Rare	Testicular disorder*, Prostatitis, Breast disorder female, Epididymal tenderness, Epididymitis, Pelvic pain, Vulval ulceration
Congenital, familial and genetic disorders	Rare	Aplasia, Gastrointestinal malformation, Ichthyosis
General disorders and administration site conditions	Very Common	Pyrexia*, Fatigue, Asthenia
	Common	Oedema (inc peripheral), Chills, Pain*, Malaise*
	Uncommon	General physical health deterioration*, Face oedema*, Injection site reaction*, Mucosal disorder*, Chest pain, Gait disturbance, Feeling cold, Extravasation*, Catheter related complication*, Change in thirst*, Chest discomfort, Feeling of body temperature change*, Injection site pain*
	Rare	Death (inc sudden), Multi-organ failure, Injection site haemorrhage*, Hernia(inc hiatus)*, Impaired healing*, Inflammation, Injection site phlebitis*, Tenderness, Ulcer, Irritability, Non-cardiac chest pain, Catheter site pain, Sensation of foreign body
	Common	Weight decreased
	Uncommon	Hyperbilirubinaemia*, Protein analyses abnormal*, Weight increased, Blood test abnormal*, C-reactive protein increased
	Rare	Blood gases abnormal*, Electrocardiogram abnormalities (inc QT prolongation)*, International normalised ratio abnormal*, Gastric pH decreased, Platelet aggregation increased, Troponin I increased, Virus identification and serology*, Urine analysis abnormal*
	Uncommon	Fall, Contusion
Injury, poisoning and procedural complications	Rare	Transfusion reaction, Fractures*, Rigors*, Face injury, Joint injury*, Burns, Laceration, Procedural pain, Radiation injuries*
	Rare	Macrophage activation
Surgical and medical procedures	Rare	

NOS=not otherwise specified
*Grouping of more than one MedDRA preferred term. # Postmarketing adverse reaction
Mantle Cell Lymphoma (MCL)
The safety profile of bortezomib in patients treated with rituximab, cyclophosphamide, doxorubicin, and prednisone (Bortezomib-R- CAP) versus patients treated with rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisone (R-CHOP) was relatively consistent to that observed in patients with multiple myeloma with main differences described below. Additional adverse drug reactions identified associated with the use of the combination therapy (Bortezomib-R-CAP) were hepatitis B infection and myocardial ischaemia. The similar incidences of these events in both treatment arms, indicated that these adverse drug reactions are not attributable to bortezomib alone. Very common incidence of the haematological adverse reactions (neutropenia, thrombocytopenia, leukopenia, anemia, lymphopenia), peripheral sensory neuropathy, hypertension, pyrexia, pneumonia, stomatitis, and hair disorders were seen with patients in the multiple myeloma
Adverse drug reactions identified as those with common incidence, similar or higher incidence in the bortezomib-R-CAP arm and with at least a possible or probable causal relationship to the components of the bortezomib-R-CAP arm, are listed in Table 8 below. Also included are adverse drug reactions identified in the bortezomib-R-CAP arm that were considered by investigators to have at least a possible or probable causal relationship to bortezomib based on historical data in the multiple myeloma studies.
Adverse reactions are listed below by system organ class and frequency grouping. Frequencies are defined as: Very common; common; uncommon; rare; very rare; not known (cannot be estimated from the available data). Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness. Table 6 has been generated using Version 16 of the MedDRA.

Table 6 Adverse reactions in patients with Mantle Cell Lymphoma treated with bortezomib-R-CAP		
System Organ Class	Incidence	Adverse reaction
Infections and infestations	Very Common	Pneumonia*
	Common	Sepsis (inc septic shock)*, Herpes zoster (inc disseminated & ophthalmic), Herpes virus infection*, Bacterial infections*, Upper/lower respiratory tract infection*, Fungal infection*, Herpes simplex*
	Uncommon	Hepatitis B, Infection*, Bronchopneumonia
Blood and lymphatic system disorders	Very Common	Thrombocytopenia*, Febrile neutropenia, Neutropenia*, Leukopenia*, Anaemia*, Lymphopenia*
	Uncommon	Pancytopenia*
Immune system disorders	Common	Hypersensitivity*
	Uncommon	Anaphylactic reaction
Metabolism and nutrition disorders	Very Common	Decreased appetite
	Common	Hypokalaemia*, Blood glucose abnormal*, Hyponatraemia*, Diabetes mellitus*, Fluid retention
	Uncommon	Tumour lysis syndrome
Psychiatric disorders	Common	Sleep disorders and disturbances*
Nervous system disorders	Very Common	Peripheral sensory neuropathy, Dysaesthesia*, Neuralgia*
	Common	Neuropathies*, Motor neuropathy*, Loss of consciousness (inc syncope), Encephalopathy*, Peripheral sensorimotor neuropathy, Dizziness*, Dysgeusia*, Autonomic neuropathy
	Uncommon	Autonomic nervous system imbalance
Eye disorders	Common	Vision abnormal*
Ear and labyrinth disorders	Common	Dysacusis (inc tinnitus)*
	Uncommon	Vertigo*, Hearing impaired (up to and inc deafness)
Cardiac disorders	Common	Cardiac fibrillation (inc atrial), Arrhythmia*, Cardiac failure (inc left and right ventricular)*, Myocardial ischaemia, Ventricular dysfunction*
	Uncommon	Cardiovascular disorder (inc cardiogenic shock)
Vascular disorders	Common	Hypertension*, Hypotension*, Orthostatic hypotension
	Common	Dyspnoea*, Cough*, Hiccups
Respiratory, thoracic and mediastinal disorders	Uncommon	Acute respiratory distress syndrome, Pulmonary embolism, Pneumonitis, Pulmonary hypertension, Pulmonary oedema (inc acute)
	Very Common	Nausea and vomiting symptoms*, Diarrhoea*, Stomatitis*, Constipation
Gastrointestinal disorders	Common	Gastrointestinal haemorrhage (inc mucosal)*, Abdominal distension, Dyspepsia, Oropharyngeal pain*, Gastritis*, Oral ulceration*, Abdominal discomfort, Dysphagia, Gastrointestinal inflammation*, Abdominal pain (inc gastrointestinal and splenic pain)*, Oral disorder*
	Uncommon	Colitis (inc clostridium difficile)*
	Common	Hepatotoxicity (inc liver disorder)
Hepatobiliary disorders	Common	Hepatic failure
	Very Common	Hair disorder*
Skin and subcutaneous tissue disorders	Common	Pruritus*, Dermatitis*, Rash*
Musculoskeletal and connective tissue disorders	Common	Muscle spasms*, Musculoskeletal pain*, Pain in extremity
Renal and urinary disorders	Common	Urinary tract infection*
General disorders and administration site conditions	Very Common	Pyrexia*, Fatigue, Asthenia
	Common	Oedema (inc peripheral), Chills, Injection site reaction*, Malaise*
	Common	Hyperbilirubinaemia*, Protein analyses abnormal*, Weight decreased, Weight increased

* Grouping of more than one MedDRA preferred term.
Description of selected adverse reactions
Retreatment of patients with relapsed multiple myeloma
Bortezomib retreatment was administered in patients with relapsed multiple myeloma, who previously had at least partial response on a bortezomib-containing regimen, the most common all-grade adverse events occurring in patients were thrombocytopenia, neuropathy, anaemia, diarrhoea, and constipation. All grade peripheral neuropathy and grade ≥ 3 peripheral neuropathy were observed.
Overdosage
In patients, overdose more than twice the recommended dose has been associated with the acute onset of symptomatic hypotension and thrombocytopenia with fatal outcomes. There is no known specific antidote for bortezomib overdose. In the event of an overdose, the patient's vital signs should be monitored and appropriate supportive care given to maintain blood pressure (such as fluids, pressors, and/or inotropic agents) and body temperature.

PHARMACOLOGICAL PROPERTIES
Mechanism of action
Bortezomib is a proteasome inhibitor. It is specifically designed to inhibit the chymotrypsin-like activity of the 26S proteasome in mammalian cells. The 26S proteasome is a large protein complex that degrades ubiquitinated proteins. The ubiquitin-proteasome pathway plays an essential role in regulating the turnover of specific proteins, thereby maintaining homeostasis within cells. Inhibition of the 26S proteasome prevents this targeted proteolysis and affects multiple signalling cascades within the cell, ultimately resulting in cancer cell death. Bortezomib is highly selective for the proteasome. At 10 µM concentrations, bortezomib does not inhibit any of a wide variety of receptors and proteases screened and is more than 1500-fold more selective for the proteasome than for its next preferable enzyme. The kinetics of proteasome inhibition were evaluated in vitro, and bortezomib was shown to dissociate from the proteasome with a t½ of 20 minutes, thus demonstrating that proteasome inhibition by bortezomib is reversible.
Bortezomib-mediated proteasome inhibition affects cancer cells in a number of ways, including, but not limited to, altering regulatory proteins, which control cell cycle progression and nuclear factor kappa B (NF-κB) activation. Inhibition of the proteasome results in cell cycle arrest and apoptosis. NF-κB is a transcription factor whose activation is required for many aspects of tumorigenesis, including cell growth and survival, angiogenesis, cell-cell interactions, and metastasis. In myeloma, bortezomib affects the ability of myeloma cells to interact with the bone marrow microenvironment.
Experiments have demonstrated that bortezomib is cytotoxic to a variety of cancer cell types and that cancer cells are more sensitive to the pro-apoptotic effects of proteasome inhibition than normal cells. Bortezomib causes reduction of tumour growth in vivo in many preclinical tumour models, including multiple myeloma.
Data from in vitro, ex-vivo, and animal models with bortezomib suggest that it increases osteoblast differentiation and activity and inhibits osteoclast function. These effects have been observed in patients with multiple myeloma affected by an advanced osteolytic disease and treated with bortezomib.

Pharmacokinetics Properties
Absorption:
Following intravenous bolus administration of a 1.0 mg/m² and 1.3 mg/m² dose to patients with multiple myeloma and creatinine clearance values greater than 50 ml/min, the mean first-dose maximum plasma concentrations of bortezomib were 57 and 112 ng/ml, respectively. In subsequent doses, mean maximum observed plasma concentrations ranged from 67 to 106 ng/ml for the 1.0 mg/m² dose and 89 to 120 ng/ml for the 1.3 mg/m² dose.
Following an intravenous bolus or subcutaneous injection of a 1.3 mg/m² dose to patients with multiple myeloma (in intravenous and subcutaneous group), the total systemic exposure after repeat dose administration (AUC_{0-∞}) was equivalent for subcutaneous and intravenous administrations. The C_∞ after subcutaneous administration (20.4 ng/ml) was lower than intravenous (223 ng/ml). The AUC_∞ geometric mean ratio was 0.99 and 90% confidence intervals were 80.18%-122.80%.
Distribution
The mean distribution volume (V_d) of bortezomib ranged from 1659 l to 3294 l following single- or repeated-dose administration of 1.0 mg/m² or 1.3 mg/m² to patients with multiple myeloma. This suggests that bortezomib distributes widely to peripheral tissues. Over a bortezomib concentration range of 0.01 to 1.0 µg/ml, the in vitro protein binding averaged 82.9% in human plasma. The fraction of bortezomib bound to plasma proteins was not concentration-dependent.

Biotransformation
In vitro studies with human liver microsomes and human cDNA-expressed cytochrome P450 isozymes indicate that bortezomib is primarily oxidatively metabolized via cytochrome P450 enzymes, 3A4, 2C19, and 1A2. The major metabolic pathway is deboronation to form two deboronated metabolites that subsequently undergo hydroxylation to several metabolites. Deboronated-bortezomib metabolites are inactive as 26S proteasome inhibitors.
Elimination
The mean elimination half-life (t½) of bortezomib upon multiple dosing ranged from 40-193 hours. Bortezomib is eliminated more rapidly following the first dose compared to subsequent doses. Mean total body clearances were 102 and 112 l/h following the first dose for doses of 1.0 mg/m² and 1.3 mg/m², respectively, and ranged from 15 to 32 l/h and 18 to 32 l/h following subsequent doses for doses of 1.0 mg/m² and 1.3 mg/m², respectively.

Special populations Hepatic impairment
The effect of hepatic impairment on the pharmacokinetics of bortezomib during the first treatment cycle, including patients primarily with solid tumors and varying degrees of hepatic impairment at bortezomib doses ranging from 0.5 to 1.3 mg/m².
When compared to patients with normal hepatic function, mild hepatic impairment did not alter dose-normalized bortezomib AUC. However, the dose-normalized mean AUC values were increased by approximately 60% in patients with moderate or severe hepatic impairment. A lower starting dose is recommended in patients with moderate or severe hepatic impairment, and those patients should be closely monitored.

Renal impairment
In patients with various degrees of renal impairment who were classified according to their creatinine clearance values (CrCL) into the following groups: Normal (CrCL≥60 ml/min/1.73 m²), Mild (CrCL = 40-59 ml/min/1.73 m²), Moderate (CrCL = 20-39 ml/min/1.73 m²), and Severe (CrCL < 20 ml/min/1.73 m²). Patients were administered intravenous doses of 0.7 to 1.3 mg/m² of bortezomib twice weekly. Exposure of bortezomib (dose-normalized AUC and Cmax) was comparable among all the groups.

Age
The pharmacokinetics of bortezomib were characterized following twice weekly intravenous bolus administration of 1.3 mg/m² doses to pediatric patients (2-16 years old) with acute lymphoblastic leukemia (ALL) or acute myeloid leukemia (AML). Based on a population pharmacokinetic analysis, clearance of bortezomib increased with increasing body surface area (BSA). Geometric mean (90% CV) clearance was 7.19 l/h/m², volume of distribution at steady-state was 834 L/m², and the elimination half-life was 100 hours. After correcting for the BSA effect, other demographics such as age, body weight and sex did not have clinically significant effects on bortezomib clearance. BSA-normalized clearance of bortezomib in pediatric patients was similar to that observed in adults.

PHARMACEUTICAL INFORMATION
Keep all medicines out of reach of children.
Storage:
Store below 30°C. Protect from light.
Shelf Life:
Unopened Vials: 30 months, Protect from light
Reconstituted Solution:
When reconstituted as directed, the reconstituted solution may be stored for 8 hours at 25°C in the original vial or a syringe prior to administration. ; however the total storage time for the reconstituted medicinal product should not exceed 8 hours prior to administration. At 2-8°C reconstituted drug product is stable for 15 days
Dosage Forms and packaging available Dosage Form: Lyophilized Powder for injection Packaging: Filled vial packed with basecap and sleeve. One vial in carton.

Manufactured by:
Dr. REDDY'S LABORATORIES LTD.,
Formulation Unit VII,
Plot Nos. P1 to P9 Phase III, Duvvada, VSEZ, Visakhapatnam,
Andhra Pradesh- 530 046, INDIA.

PRODUCT REGISTRATION HOLDER
Dr. Reddy's Laboratories Malaysia Sdn. Bhd.
UNIT NO. SO-29-07 AND SO-29-08, MENARA 1,
STRATA OFFICE, NO. 3, JALAN BANGSAR, KLECO CITY,
59200 KUALA LUMPUR
MALAYSIA
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